

ID627001: Introductory Game Development

Portfolio Marking Rubric

Functionality - Learning Outcome 1 (55%)

	A – 10-8	B – 7.5-6.5	C – 6-5	D/E – 4.5-0
Pong Game (10%)	Game fully published on itch.io with all fundamental mechanics implemented to a high standard. Main menu is professional, intuitive and fully functional. Paddle movement, ball movement and collisions work flawlessly. Score system and display work perfectly with persistent high scores. Pause/resume functionality works seamlessly with the P key. Sound effects are appropriate and enhance gameplay. Game over condition (score of 10) and end game menu work perfectly. All 3 selected Tier 2 mechanics implemented to a high standard. All 2 selected Tier 3 mechanics implemented to a high standard. All 3 custom mechanics are original, well-implemented and enhance gameplay.	Game published on itch.io with most fundamental mechanics implemented effectively. Main menu works with minimal issues. Core gameplay (paddle/ball movement, collisions) works well with minor inconsistencies. Score system and display work with minor issues. Pause/resume and sound effects implemented with small flaws. Game over and end game menu function as intended with minimal issues. All 3 Tier 2 mechanics implemented with minor flaws. Both Tier 3 mechanics implemented with some issues. All 3 custom mechanics implemented but may lack polish or originality.	Game published on itch.io but with incomplete implementation or notable bugs. Main menu exists but has functional issues. Core gameplay mechanics function but with obvious flaws. Score system works but may have persistence issues. Pause/resume exists but may be buggy. Sound effects present but limited or inappropriate. Game over condition works but menu may be basic or have issues. 2-3 Tier 2 mechanics implemented but with significant issues. At least 1 Tier 3 mechanic implemented. At least 2 custom mechanics implemented but poorly executed.	Game partially published or not published on itch.io. Main menu missing or severely broken. Core mechanics partially implemented or severely broken. Score system missing or non-functional. Pause/resume missing or non-functional. Sound effects missing or inappropriate. Game over condition unreliable or missing. Less than 2 Tier 2 mechanics implemented. No Tier 3 mechanics implemented. Less than 2 custom mechanics or mechanics that do not function.
	A – 15-12	B – 11.5-10	C – 9.5-7.5	D/E – 7-0

Breakout Game (15%)	Game fully published on itch.io with all fundamental mechanics implemented to a high standard. Main menu is professional, intuitive and fully functional. Paddle movement, ball movement and brick collisions work flawlessly. 10x4 brick grid with health system perfectly implemented. Score system and 3-life system work perfectly with display. High scores system stores and displays top 10 scores accurately. Sound effects are appropriate and enhance gameplay. Pause/resume functionality works seamlessly with P key. Game over conditions work perfectly (all bricks destroyed or all lives lost). End game menu works perfectly. All 3 selected Tier 2 mechanics implemented to a high standard. Both selected Tier 3 mechanics implemented to a high standard. All 3 custom mechanics are original, well-implemented and enhance gameplay.	Game published on itch.io with most fundamental mechanics implemented effectively. Main menu works with minimal issues. Core gameplay mechanics work well with minor inconsistencies. Brick grid and health system work well with minor issues. Lives and score systems function with minor display issues. High scores work with occasional inconsistencies. Sound effects implemented with small flaws. Pause/resume works with minor issues. Game over conditions function as intended. End game menu works with minimal issues. All 3 Tier 2 mechanics implemented with minor flaws. Both Tier 3 mechanics implemented with some issues. All 3 custom mechanics implemented but may lack polish or originality.	Game published on itch.io but with incomplete implementation or notable bugs. Main menu exists but has functional issues. Core gameplay mechanics function but with obvious flaws. Brick grid implemented but issues with health system. Lives and score systems work but may have display or tracking issues. High scores system functions but with significant limitations. Sound effects present but limited or inappropriate. Pause/resume exists but may be buggy. Game over conditions implemented but not always reliable. End game menu basic or with issues. 2-3 Tier 2 mechanics implemented but with significant issues. At least 1 Tier 3 mechanic implemented. At least 2 custom mechanics implemented but poorly executed.	Game partially published or not published on itch.io. Main menu missing or severely broken. Core mechanics partially implemented or severely broken. Brick grid incomplete or severely flawed. Lives or score systems missing or non-functional. High scores missing or non-functional. Sound effects missing or inappropriate. Pause/resume missing or non-functional. Game over conditions unreliable or missing. End game menu missing or non-functional. Less than 2 Tier 2 mechanics implemented. No Tier 3 mechanics implemented. Less than 2 custom mechanics or mechanics that do not function.
	A – 15-12	B – 11.5-10	C – 9.5-7.5	D/E – 7-0

Space Invaders Game (15%)	Game fully published on itch.io with all fundamental mechanics implemented to a high standard. Main menu is professional, intuitive and fully functional. Ship movement, enemy formation and movement work flawlessly. Shooting and collision detection work perfectly for all game elements. Score system and 3-life system work perfectly with display. Player name entry works with Random User Generator API fallback. High scores system stores and displays top 10 scores accurately. Sound effects are appropriate and enhance gameplay. Pause/resume functionality works seamlessly with P key. Game over conditions work perfectly. End game menu works perfectly. All 3 selected Tier 2 mechanics implemented to a high standard. Both selected Tier 3 mechanics implemented to a high standard. All 3 custom mechanics are original, well-implemented and enhance gameplay.	Game published on itch.io with most fundamental mechanics implemented effectively. Main menu works with minimal issues. Ship and enemy movement work well with minor inconsistencies. Shooting and collision detection work with occasional issues. Lives and score systems function with minor display issues. Player name entry works but API integration may have minor issues. High scores work with occasional inconsistencies. Sound effects implemented with small flaws. Pause/resume works with minor issues. Game over conditions function as intended. End game menu works with minimal issues. All 3 Tier 2 mechanics implemented with minor flaws. Both Tier 3 mechanics implemented with some issues. All 3 custom mechanics implemented but may lack polish or originality.	Game published on itch.io but with incomplete implementation or notable bugs. Main menu exists but has functional issues. Ship and enemy movement function but with obvious flaws. Shooting or collision detection implemented with significant issues. Lives and score systems work but may have display or tracking issues. Player name entry basic without API fallback or with issues. High scores system functions but with significant limitations. Sound effects present but limited or inappropriate. Pause/resume exists but may be buggy. Game over conditions implemented but not always reliable. End game menu basic or with issues. 2-3 Tier 2 mechanics implemented but with significant issues. At least 1 Tier 3 mechanic implemented. At least 2 custom mechanics implemented but poorly executed.	Game partially published or not published on itch.io. Main menu missing or severely broken. Ship or enemy movement severely flawed or incomplete. Shooting or collision detection missing or non-functional. Lives or score systems missing or non-functional. Player name entry missing. High scores missing or non-functional. Sound effects missing or inappropriate. Pause/resume missing or non-functional. Game over conditions unreliable or missing. End game menu missing or non-functional. Less than 2 Tier 2 mechanics implemented. No Tier 3 mechanics implemented. Less than 2 custom mechanics or mechanics that do not function.
	A – 15-12	B – 11.5-10	C – 9.5-7.5	D/E – 7-0

RogueLike Game (15%)	Game fully published on itch.io with all fundamental mechanics implemented to a high standard.	Game published on itch.io with most fundamental mechanics implemented effectively.	Game published on itch.io but with incomplete implementation or notable bugs.	Game partially published or not published on itch.io.
	Main menu is professional, intuitive and fully functional.	Main menu works with minimal issues.	Main menu exists but has functional issues.	Main menu missing or severely broken.
	Player and enemy movement mechanics work flawlessly.	Player and enemy movement work well with minor inconsistencies.	Player and enemy movement function but with obvious flaws.	Player or enemy movement severely flawed or incomplete.
	Combat system works perfectly with appropriate difficulty balance.	Combat system works with occasional balance issues.	Combat system implemented but with significant balance issues.	Combat system missing or non-functional.
	Health system for player and enemies works perfectly with clear display.	Health system functions with minor display issues.	Health system works but may have display or calculation issues.	Health system missing or non-functional.
	Score system works perfectly with display.	Score system works with minor issues.	Score system works but may have tracking issues.	Score system missing or non-functional.
	Pickup system is varied and enhances gameplay.	Pickup system works but may lack variety.	Basic pickup system implemented.	Pickup system missing or severely limited.
	High scores system stores and displays top 10 scores accurately.	High scores work with occasional inconsistencies.	High scores system functions but with significant limitations.	High scores missing or non-functional.
	Sound effects are appropriate and enhance gameplay.	Sound effects implemented with small flaws.	Sound effects present but limited or inappropriate.	Sound effects missing or inappropriate.
	Pause/resume functionality works seamlessly with P key.	Pause/resume works with minor issues.	Pause/resume exists but may be buggy.	Pause/resume missing or non-functional.
	Game over conditions work perfectly.	Game over conditions function as intended.	Game over conditions implemented but not always reliable.	Game over conditions unreliable or missing.
	End game menu works perfectly.	End game menu works with minimal issues.	End game menu basic or with issues.	End game menu missing or non-functional.
	All 3 selected Tier 2 mechanics implemented to a high standard.	All 3 Tier 2 mechanics implemented with minor flaws.	2-3 Tier 2 mechanics implemented but with significant issues.	Less than 2 Tier 2 mechanics implemented.
	Both selected Tier 3 mechanics implemented to a high standard.	Both Tier 3 mechanics implemented with some issues.	At least 1 Tier 3 mechanic implemented.	No Tier 3 mechanics implemented.
	All 4 custom mechanics are original, well-implemented and enhance gameplay.	All 4 custom mechanics implemented but may lack polish or originality.	At least 3 custom mechanics implemented but poorly executed.	Less than 3 custom mechanics or mechanics that do not function.
		Good group collaboration with mostly even workload distribution.	Evidence of group collaboration but uneven workload distribution.	Poor evidence of group collaboration or highly uneven workload distribution.

	Evidence of effective group collaboration and even workload distribution.			
--	---	--	--	--

Code Quality and Best Practices - Learning Outcome 1 (40%)

	A – 10-8	B – 7.5-6.5	C – 6-5	D/E – 4.5-0
Project Structure (10%)	Exceptional organisation with a clear and logical structure. Excellent separation of concerns using appropriate design patterns. Outstanding modularity promoting high reusability and maintainability. Components have single responsibilities and clear interfaces.	Good organisation with mostly clear and logical structure. Good separation of concerns with some design patterns. Good modularity supporting reusability and maintainability. Most components have single responsibilities.	Adequate organization with somewhat clear structure. Basic separation of concerns. Some modularity supporting limited reusability. Some components have multiple responsibilities.	Poor organization with unclear or illogical structure. Little to no separation of concerns. Limited modularity with poor reusability and maintainability. Many components have multiple responsibilities
	A – 5-4	B – 3.5	C – 3-2.5	D/E – 2-0
Naming Conventions (5%)	Exemplary naming that is consistently clear, descriptive, and meaningful. Perfect adherence to C# naming conventions (PascalCase, camelCase, etc.). Names accurately convey purpose and behaviour throughout codebase.	Very good naming that is mostly clear and descriptive. Good adherence to C# naming conventions with minor inconsistencies. Names generally convey purpose and behaviour.	Adequate naming that is sometimes unclear or non-descriptive. Inconsistent adherence to C# naming conventions. Some names fail to convey purpose or behaviour.	Poor naming that is frequently unclear, cryptic, or misleading. Little adherence to C# naming conventions. Names fail to convey purpose or behaviour
	A – 5-4	B – 3.5	C – 3-2.5	D/E – 2-0

Documentation and Comments (5%)	Comprehensive XML documentation for all public members. Excellent commenting for complex logic and algorithms. Documentation includes parameter descriptions and return values. Comments explain "why" not just "what" where appropriate	Good XML documentation for most public members. Good commenting for complex sections. Documentation includes most parameter descriptions and return values. Comments generally helpful in understanding the code.	Basic XML documentation for some public members. Limited commenting for complex sections. Documentation often missing parameter descriptions or return values. Comments sometimes unclear or unhelpful.	Little to no XML documentation. Missing comments for complex sections. Documentation missing parameter descriptions and return values. Comments absent, unclear, or misleading
	A – 5-4	B – 3.5	C – 3-2.5	D/E – 2-0
Code Formatting (5%)	Flawless and consistent indentation, spacing, and bracing. Excellent use of blank lines to separate logical sections. Consistent line length and formatting across entire codebase. Code appears to follow a style guide with absolute consistency.	Good and mostly consistent indentation, spacing, and bracing. Good use of blank lines to separate logical sections. Generally consistent line length and formatting. Code mostly follows a style guide.	Inconsistent indentation, spacing, or bracing. Limited use of blank lines between logical sections. Inconsistent line length and formatting. Code inconsistently follows a style guide.	Poor indentation, spacing, and bracing practices. No clear use of blank lines between logical sections. Inconsistent and difficult-to-read formatting. No evidence of following a style guide.
	A – 5-4	B – 3.5	C – 3-2.5	D/E – 2-0

Dead Code (5%)	No unused or redundant code elements anywhere in the project. Clean codebase with no commented-out code blocks. No debug statements or temporary code left in production code.	Very few instances of unused or redundant code elements. Mostly clean codebase with minimal commented-out code. Few debug statements or temporary code remaining.	Several instances of unused or redundant code elements. Some commented-out code blocks remain. Some debug statements or temporary code left in production code.	Numerous instances of unused or redundant code elements. Many commented-out code blocks throughout the codebase. Debug statements or temporary code frequently left in production code.
	A – 10-8	B – 7.5-6.5	C – 6-5	D/E – 4.5-0
Performance and Scalability (10%)	Highly optimized code using appropriate algorithms and data structures. Excellent resource management (memory, CPU, etc.). Thoughtful handling of potential performance bottlenecks. Code structured to handle scaling with additional features/content.	Generally optimized code with appropriate algorithms. Good resource management with few inefficiencies. Most potential performance bottlenecks addressed. Code can handle some scaling with additional features.	Some inefficient code with inappropriate algorithms in places. Basic resource management with some inefficiencies. Some performance bottlenecks not addressed. Code would struggle with scaling.	Poor resource management with significant inefficiencies. Performance bottlenecks ignored or exacerbated. Code cannot handle any scaling.

Documentation and Git Usage - Learning Outcome 1 (5%)

	A – 5-4	B – 3.5	C – 3-2.5	D/E – 2-0
GitHub Issues and Repository Documentation (2.5%)	Exceptional and consistent use of GitHub issues throughout the entire development process. Issues are well-organised, properly labelled, and contain detailed descriptions. Progress is clearly tracked with regular updates and appropriate issue closure. Comprehensive README with all URLs to the applications on itch.io properly documented. Documentation is exceptionally well-organised and easy to follow. Expert use of Markdown formatting with appropriate headings, lists, tables, code blocks, links, and emphasis. Documentation is visually appealing and highly readable. Flawless spelling and grammar throughout all documentation.	Good and generally consistent use of GitHub issues throughout development. Issues are organised with appropriate labels and descriptions. Progress is tracked with regular updates. Good README containing most required URLs with clear documentation. Minor omissions or areas that could be improved in clarity. Good use of Markdown formatting with appropriate structure. Some advanced formatting features utilised. Good spelling and grammar with only minor errors that do not impact understanding.	Basic use of GitHub issues but with inconsistent application. Issues have minimal descriptions and limited organisation. Progress tracking is present but sporadic. Basic README containing some required URLs but with significant omissions or unclear documentation. Basic use of Markdown with simple formatting. Limited use of advanced features. Multiple spelling and grammar errors that occasionally impact clarity but do not prevent understanding.	Minimal or inadequate use of GitHub issues. Many issues lack descriptions or organisation. Little to no evidence of progress tracking. Poor or incomplete README missing multiple required elements. Instructions are unclear, incorrect, or missing entirely. Poor use of Markdown with little to no formatting. Documentation is difficult to read due to formatting issues. Numerous spelling and grammar errors that significantly impact clarity and understanding.
	A – 5-4	B – 3.5	C – 3-2.5	D/E – 2-0

Git Configuration and Commit Practices (2.5%)	Perfect implementation of the specified .gitignore file from GitHub. No unnecessary files are tracked, and all appropriate files are ignored. Repository is free from any IDE configurations, build artifacts, or environment files. Exceptional commit messages that perfectly reflect the context of each functional requirement change. Messages follow a consistent and appropriate naming convention style (e.g., conventional commits). Commit history tells a clear story of project development with logical progression. Commits are appropriately sized with related changes grouped together.	Good implementation of the specified .gitignore file with minor oversights. Few unnecessarily tracked files. Repository is generally clean from system or build files. Good commit messages that generally reflect functional changes. Messages follow a naming convention with occasional inconsistencies. Commit history shows good progression of development. Commit sizes are generally appropriate.	Basic implementation of the specified .gitignore file with some significant oversights. Some unnecessary files are tracked. Repository contains some files that should be ignored. Basic commit messages that sometimes reflect functional changes. Inconsistent application of naming conventions. Commit history provides basic understanding of development progression. Commit sizes vary widely with some too large or too small.	Poor implementation of the specified .gitignore file or no custom .gitignore used. Many unnecessary files are tracked. Repository is cluttered with system or build files. Poor commit messages that rarely reflect functional changes. No consistent naming convention. Commit history does not clearly show development progression. Commits are inappropriately sized (too large or too frequent with minimal changes).
---	--	---	--	--

ID627001: Introductory Game Development

Portfolio Marking Cover Sheet

Name:

Date:

Learner ID:

Assessor's Name:

Assessor's Signature:

Criteria	Weighting	Mark
Functionality	55	
Code Quality and Best Practices	40	
Documentation and Git Usage	5	
Final Result	/100	
This assessment is worth 100% of the final mark for the Introductory Game Development course.		

Feedback:

Functionality:

Code Quality and Best Practices:

Documentation and Git Usage: